

* UART (Universal Asynchronous Receiver Transmitter)

* USART (Universal Synchronous Asynchronous Receiver Transmitter)

* USART Hardware Components :-

- ↳ Baud rate generator
- ↳ Tx and Rx Shift Registers
- ↳ Transmit/Receive control blocks
- ↳ Transmit/Receive buffers
- ↳ First-In, first-out (FIFO) buffer memory

NOTE: STM32F4xx
based MCUs
don't support
FIFOS.

* Summary:

↳ USART is a hardware unit in microcontroller chip which transmits and receive data bits either in Synchronous mode or in Asynchronous mode.

↳ If it is Asynchronous mode, then clock will not be sent along with the data, instead we use start and stop bits along with the useful data.

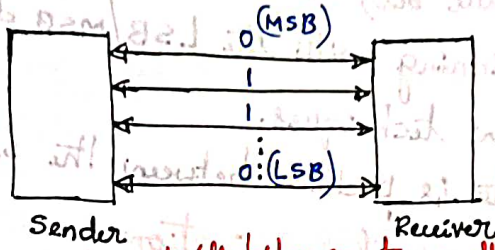
* STM ARM chips come with many on-chip USARTs, and they are designated as USART x where $x=1, 2, 3, \dots$ and so on.

* LIS3DSH: MEMS Motion Sensor: 3-axis nano accelerometer made by ST Microelectronics.

* Serial port (USART) :

↳ We have two types of communications.

1) parallel communication



* All bits are transmitted simultaneously.

NOTE:

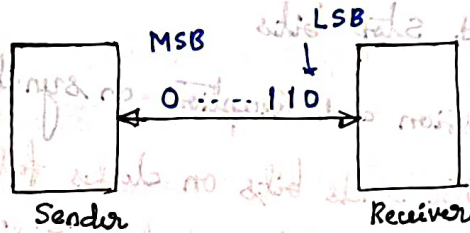
Examples:

↳ Exchange of Information between two persons

↳ Exchange of Information between two satellites

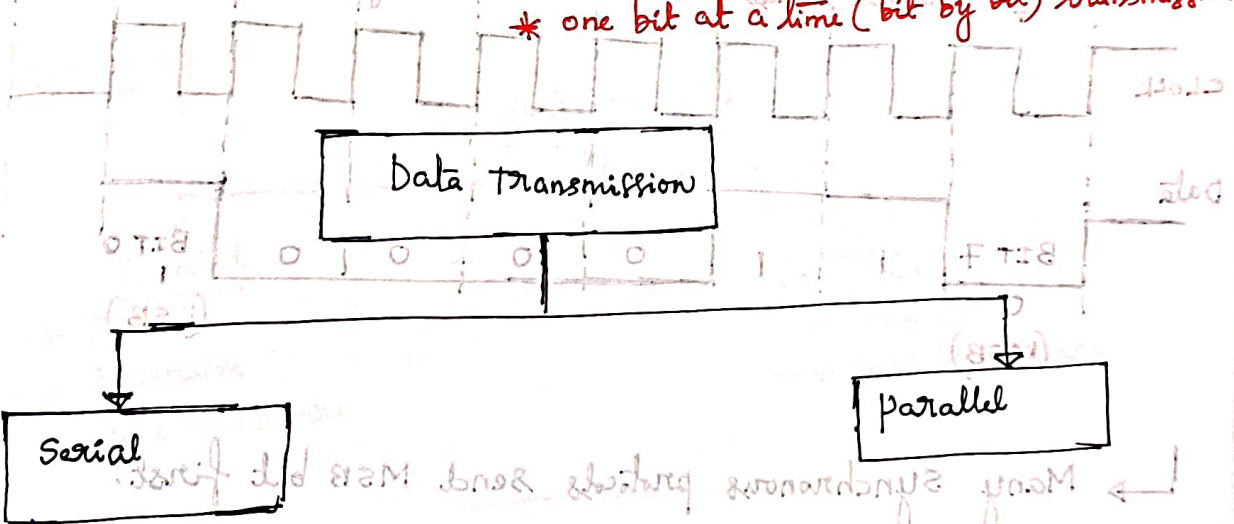
↳ Exchange of Information - action b/n two antennas.

2) Serial communication



↳ Exchange of Information - action b/n up/sec/cpu and peripherals.

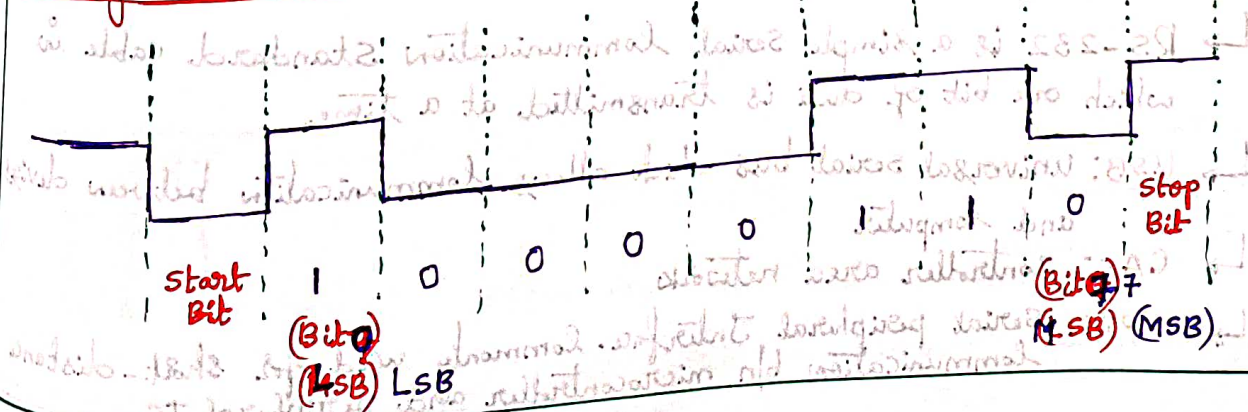
* one bit at a time (bit by bit) transmission



→ Synchronous

→ Asynchronous

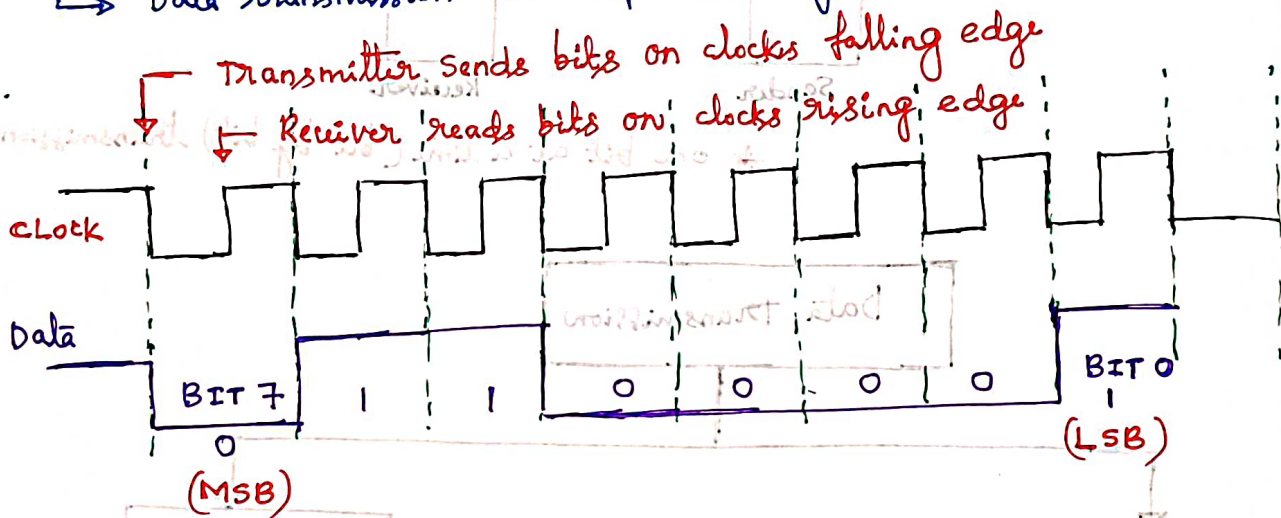
* Asynchronous serial communication :



- ↳ Asynchronous protocols send LSB first.
- ↳ uses start bit and stop bit concepts.
- ↳ After the start bit, the transmitter sends the data bits in sequence beginning with the LSB/MSB depending on little endian & big endian techniques.
- ↳ Framing: Data is packed between the start and stop bits.

* Synchronous Serial Communication:

- ↳ No start and stop bits
- ↳ Data transmission and Reception on synchronisation with clock.



- ↳ Many synchronous protocols send MSB bit first.

* Examples of serial communication protocols

- ↳ RS 232, RS 422, RS 485, USB, CAN, SPI and I2C
- ↳ RS Stands for "Recommended Standard"
- ↳ RS-232 is a simple serial communication standard cable in which one bit of data is transmitted at a time.
- ↳ **USB**: universal serial bus which allows communication between devices and computer
- ↳ **CAN**: controller area network
- ↳ **SPI**: serial peripheral Interface. commonly used for short-distance communication b/w microcontroller and peripheral IC.

↳ I2C: (Inter-Integrated controller): used to connect low-speed devices.

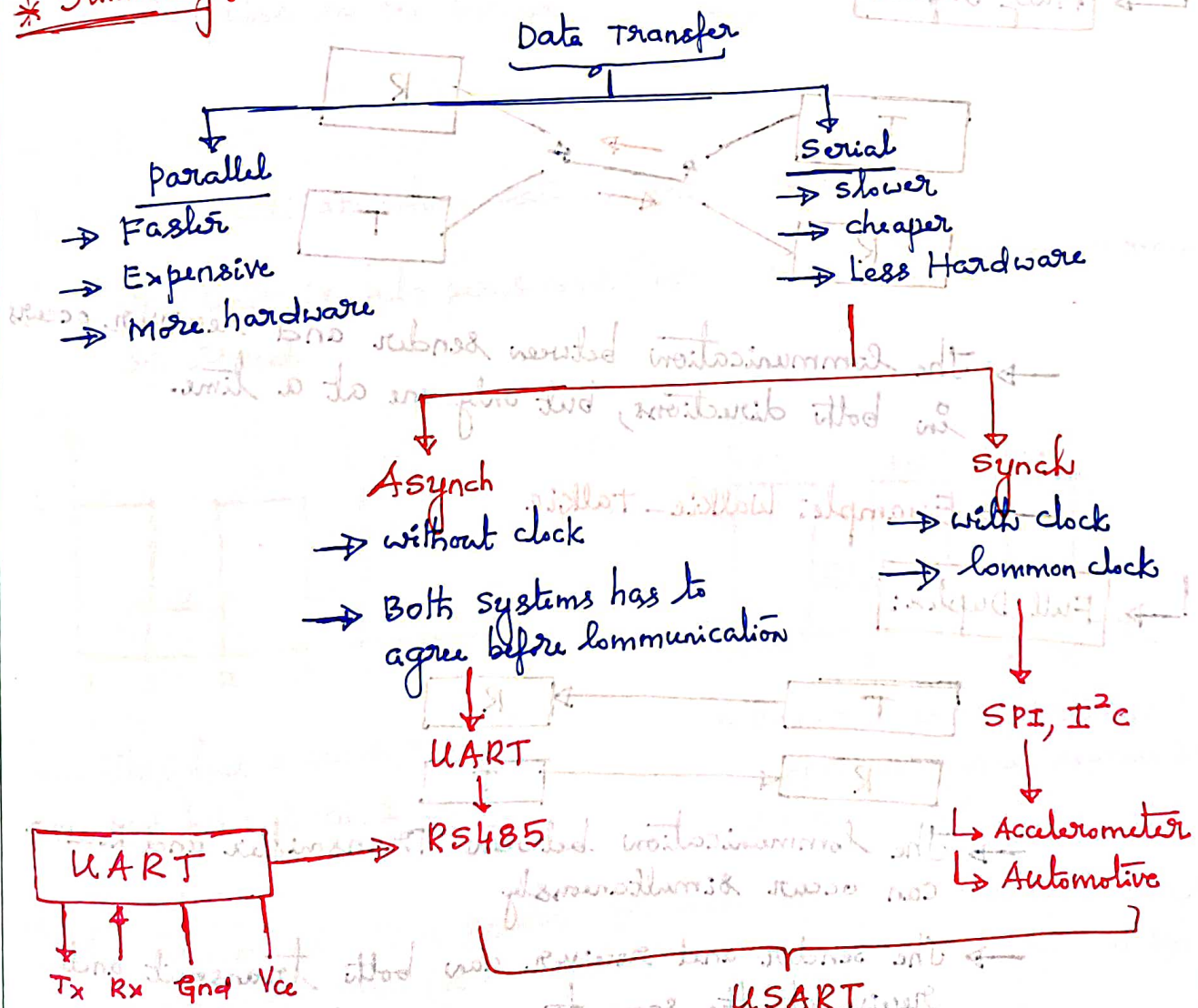
* Examples of Asynchronous serial communication protocols:
RS-232, RS-422 and RS-485.

* Example of Synchronous serial communication protocols:
I2C, SPI

* Example of Both Synchronous and Asynchronous communication protocols:
USART, USB & CAN

↳ USART: universal Synchronous/asynchronous Receiver/transmitter
It can function in a slower asynchronous mode or in a faster Synchronous mode with a clock signal.

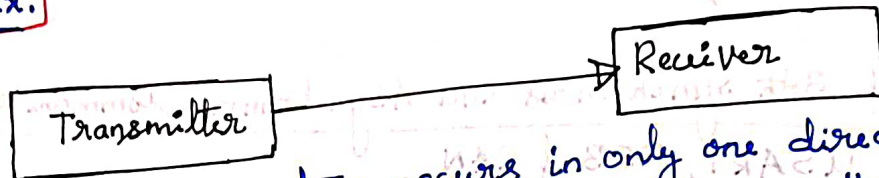
* Summary:



* Transmission modes: Simplex, Half duplex and Full duplex.

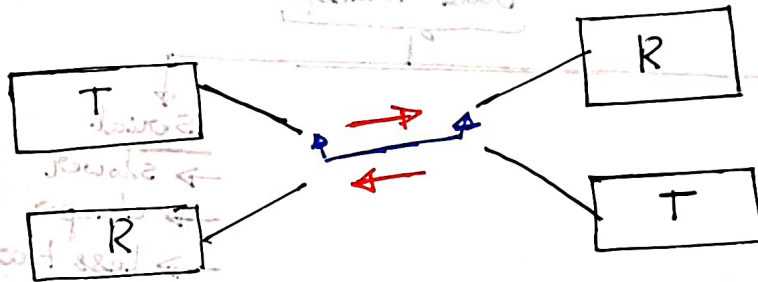
↳ Transmission mode defines the direction of data between two connected devices.

↳ Simplex:



- The communication occurs in only one direction.
- The sender can only send the data and the receiver can only receive the data.
- The receiver cannot reply to the sender.
- Example: Keyboard, loud speaker.

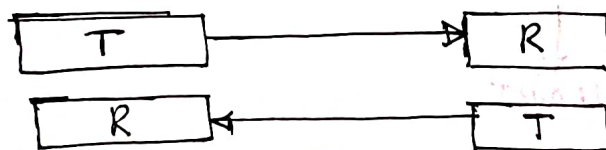
↳ Half Duplex:



- The communication between sender and receiver occurs in both directions, but only one at a time.

→ Example: Walkie-talkie

↳ Full Duplex:



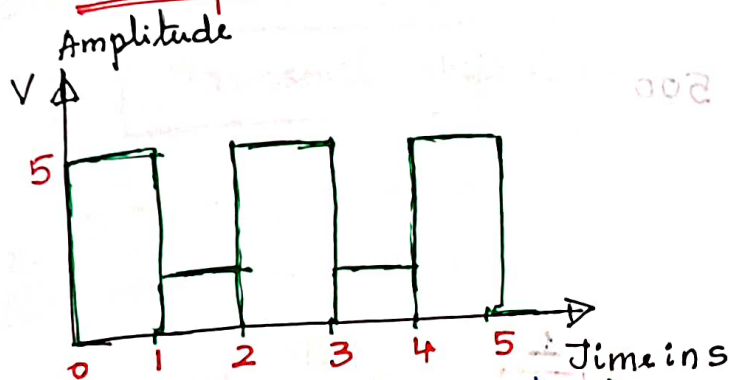
- The communication between Transmitter and Receiver can occur simultaneously.

→ The sender and receiver can both transmit and receive at the same time. Example: Telephone.

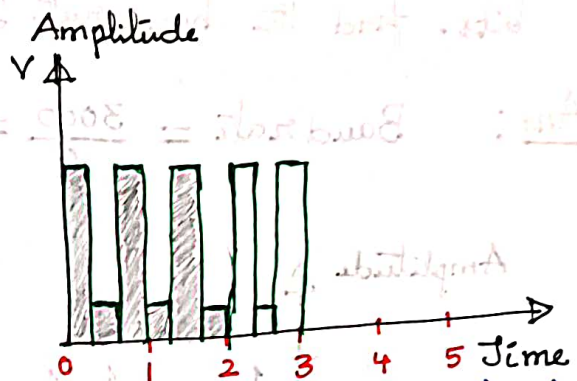
* Baud rate and Rate of data transfer (bps: bits per second):

- ↳ Baud rate is defined as the number of signal changes per second.
- ↳ Is the rate at which signals can change (pulses/second).
- ↳ It is a measurement of speed.

Examples



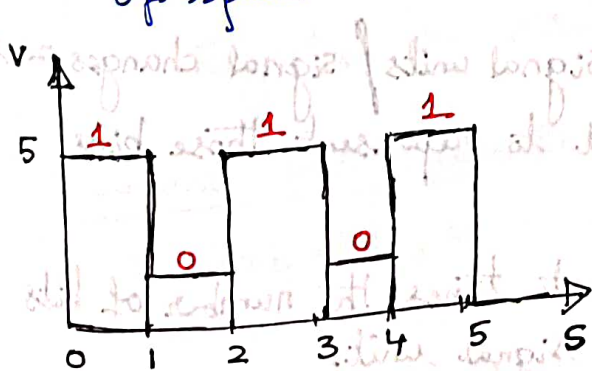
- This has a baud rate of 1
- one pulse in one second



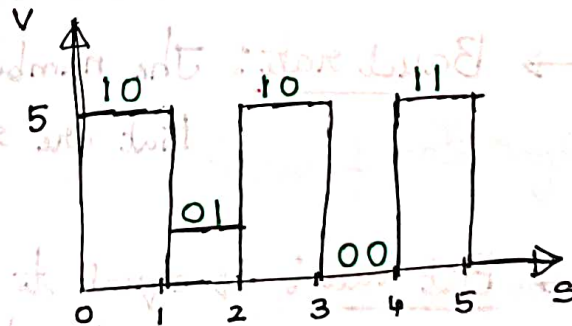
- This has a baud rate of 3
- Three pulses in one second

* Bit Rate:

- ↳ Is the rate at which data is sent (bits/second).
- ↳ The unit is bits per second (bps) and it is also a measurement of speed.



- ↳ This has a baud rate of 1 and bit rate of 1 bps



- Baud rate is 1 and bit rate is 2 bps. Each signal represents 2 bits.

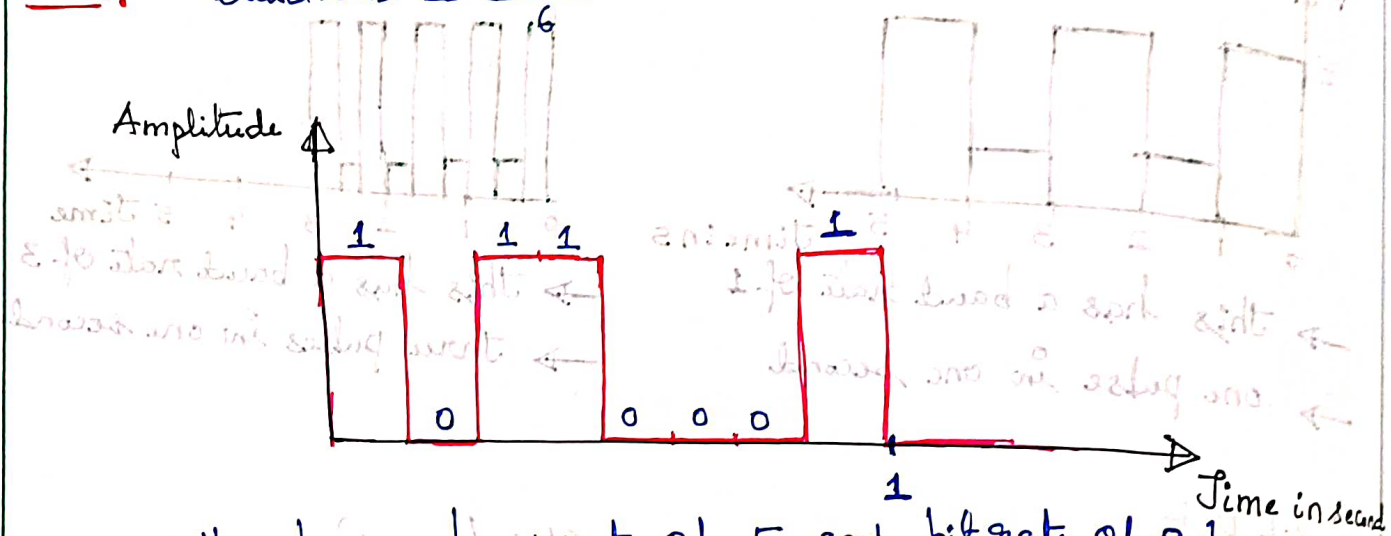
NOTE: In modems, it is possible for each signal to transfer multiple bits of data. For unmodulated signal, the baud rate and bps

P> Analog signal carries 4 bits in each signal unit. If 1000 signal units are sent/second, find the baud rate & bit rate?

Ans: Baud rate = 1000
 Bit rate = $1000 \times 4 = 4000 \text{ bps}$

P> The bit rate of a signal is 3000. Each signal unit carries 6 bits. find the baud rate?

Ans: Baud rate = $\frac{3000}{6} = 500$



→ This has a baud rate of 5 and bit rate of 8 bps.

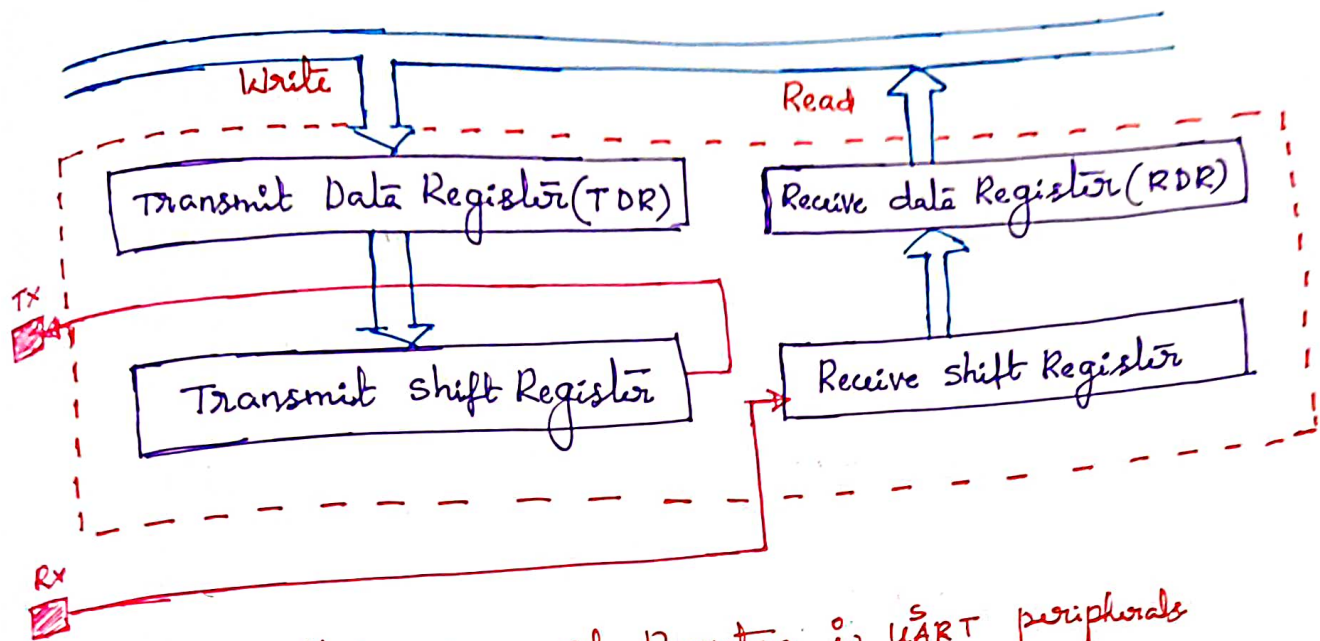
NOTE:

→ Baud rate: The number of signal units / signal changes per second that are required to represent those bits.

→ Bit rate: is equal to baud rate times the number of bits represented by each signal unit.

∴ Bit rate ≥ Baud rate

* Simplified Block diagram of USART :-



→ There are three groups of Registers in USART peripherals

1) Configuration Registers:

- Before using the USART peripheral the configuration Registers must be initialized.
- Configuration parameters are Baud rate, Word length, stop bit and Interrupts (if needed).
- In STM ARM microcontroller, some of the configuration Registers are BAUD, control 1 and control 2

2) Transmit and Receive Register:

- To send data, we simply write to the Transmit data register (USART-TDR)
- The content of Transmit data Register (TDR) is sent out through the serial transmit pin (Tx0).
- The received data is stored into the Receive data register (USART-RDR) through the serial Receive pin (Rx0).
- Both TDR and RDR have two different addresses.
- Write to the transmit register and read from the Receive register.